

Analyzing Circuits

An electrical mathematical toolkit

When tackling any complex circuit problem we should start with Kirchhoff's circuit laws and these four key electrical relationships:

$$I = \frac{\Delta Q}{\Delta t}$$

$$V = \frac{W}{Q}$$

$$P = VI$$

$$R = \frac{V}{I}$$

DIAL

An additional tool to help with analyzing circuits is the acronym **DIAL**.

DIAL stands for

D : Draw Diagrams, Label Everything

I : Identify relevant laws & principles

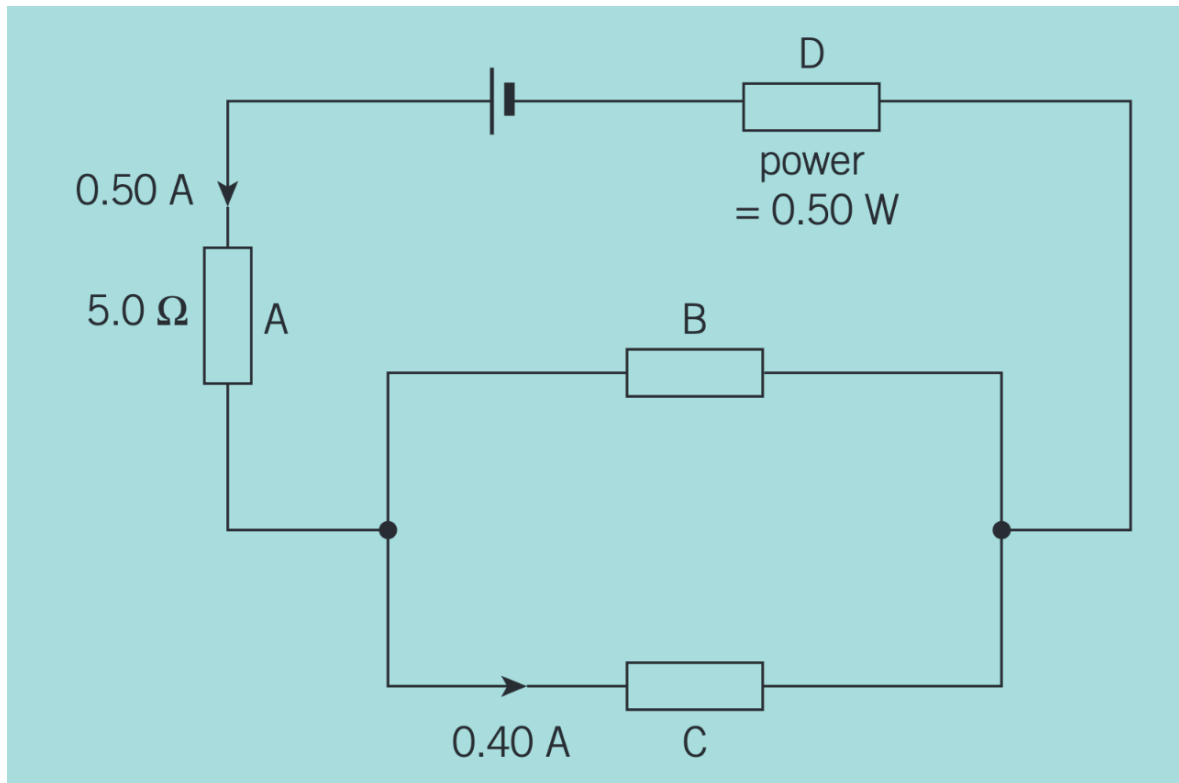
A : Apply I (general principles) to D (scenario-specific cases)

L : soLve and evaluate.

Summary Questions

1. a. *I*: current
 ΔQ : change in charge
 Δt : time elapsed
 - b. *V*: voltage
W: energy
Q: charge
 - c. *P*: power
V: voltage
I: current
 - d. *V*: voltage
I: current
R: resistance
2. a. $Q = I\Delta t$
 $= 0.3\text{A} \times 45\text{s}$
 $= 13.5\text{C}$
 - b. 2 minutes is 120 seconds.
 $W = VQ$
 $= VIt$
 $= 2.7 \times 0.075 \times 120$
 $= 24.3\text{J}$

3.



- a. From Kirchhoff's first law, the current in the branch with resistor B is 0.10A.
- b. In resistor A, the p.d. can be found by

$$\begin{aligned} V &= IR \\ &= 0.5 \times 5 \\ &= 2.5V \end{aligned}$$

The p.d. in resistor D can be found by

$$\begin{aligned} P &= IV \\ V &= \frac{P}{I} \\ &= \frac{0.5}{0.5} \\ &= 1V \end{aligned}$$

- c. In resistor D, the resistance can be found by

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{1}{0.5} \\ &= 2\Omega \end{aligned}$$

4. a. Resistor A:

$$\begin{aligned} P &= IV \\ &= 0.5 \times 2.5 \\ &= 1.25 \\ &= 1.3 \text{ W to 2 s.f.} \end{aligned}$$

b. Resistor B:

$$\begin{aligned}V &= IR \\&= 0.1 \times 8 \\&= 0.8\text{V}\end{aligned}$$

and

$$\begin{aligned}P &= IV \\&= 0.1 \times 0.8 \\&= 0.080\text{W}\end{aligned}$$

c. The voltage in resistor B = the voltage in resistor C.

Resistor C:

$$\begin{aligned}P &= IV \\&= 0.4 \times 0.8 = 0.32\text{W}\end{aligned}$$

d. Resistor D has a power of 0.50W.

The e.m.f. of the cell can be found by summing the voltages drawn in the circuit.

$$\begin{aligned}\varepsilon &= (0.50\text{A} \times 5\Omega) + (0.1\text{A} \times 8\Omega) + 1\text{V} \\ \varepsilon &= 4.3\text{V}\end{aligned}$$